

Robust In-Hand Manipulation via Priors from Grasp Quality and Contact Geometry

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Abstract—In-hand manipulation without external sensing is challenging due to uncertainties from finger-object contacts and disturbances by gravity. While reinforcement learning has shown promise in learning complex finger gaiting, existing approaches do not prioritize maintaining well-conditioned grasps for sustained manipulation. We introduce two complementary physics priors for robust in-hand rolling: a global grasp-quality prior derived from classical grasp analysis and a local contact-geometry prior based on fingertip curvature. The grasp-quality prior is used as a dense reward-shaping term that encourages well-distributed contacts with improved worst-case wrench resistance. The contact-geometry prior is expressed in the fingertip geometry that mechanically shapes the contact interface toward task-aligned rolling while reducing off-axis drift. We evaluate the effect of these priors on learning in-hand rotation for a multifingered robotic hand manipulating three objects at four palm orientations. Results show significant improvement in rotation efficiency, grasp stability, and disturbance rejection, suggesting that physics priors embedded in both learning and fingertip morphology improve task robustness. An overview video can be found at <https://youtu.be/pdd1wHxQnJM?si=dM-U5kiiPTYsk3Pk>.

I. INTRODUCTION

In-hand manipulation is an essential skill for a wide range of tasks, including regrasping, tool use, and everyday manipulation [1], [2], [3]. This process requires a set of low-level skills that unfold sequentially and concurrently: coordinating multiple fingers to guide object motion, maintaining grasp stability, and regulating finger-object contacts. Each skill is governed by task-related physics priors that directly affect the success and efficiency of manipulation. However, many current approaches treat in-hand manipulation as a generic policy-learning problem, without explicitly modeling relevant physics underlying these low-level skills [4], [5], [6], [7], [8]. In this work, we therefore investigate a central question: **To what extent can robots leverage task physics to guide and accelerate in-hand manipulation skill learning?**

To answer the questions, **we study two fundamental and complementary physics priors in in-hand rotation with a multi-fingered hand:** (1) a *computationally-imposed* global prior for grasp stability through contact coordination and (2) a *mechanically-imposed* local prior based on contact geometry, which can anisotropically alter rolling mobility at a contact. We explore how these physics priors facilitate skill acquisition in a reinforcement learning framework.

II. PROPOSED METHOD

We assume that in-hand rotation is shaped by two complementary physics priors. At the grasp level, the multi-finger

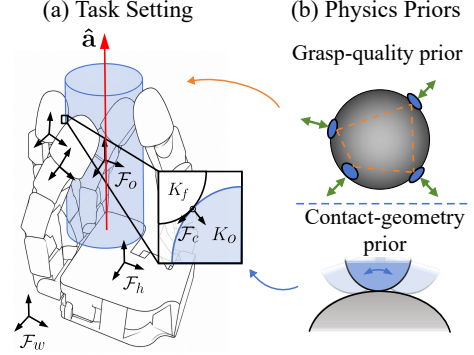


Fig. 1. We embed two physics priors, global grasp quality and local contact geometry, into the learning and fingertip design to improve in-hand rotation.

contact configuration should avoid degenerate layouts that are weak against disturbances. At the fingertip contact level, the local fingertip-object interface should permit rolling in the task direction while limiting off-axis motion. We model these two properties through a *global grasp-quality prior* derived from the grasp map and a *local contact-geometry prior* based on relative curvature. The former is used as a dense learning prior during policy training, while the latter motivates the design of anisotropic fingertip morphology.

A. Global Grasp-Quality Prior

Motivated by classical grasp analysis [9], we introduce a grasp-quality metric based on the grasp map as an inductive bias in learning, to favor well-distributed contacts and penalize uneven wrench resistance.

Consider the grasp map $\mathbf{G}$, which describes how unit contact forces generate object wrenches in multi-finger manipulation [10]. We express the contact configuration as the Gramian of the grasp map, $\mathbf{M} = \mathbf{G}\mathbf{G}^T \in \mathbb{R}^{6 \times 6}$. The eigenvectors and eigenvalues of $\mathbf{M}$ determine the principal directions and axis lengths of the wrench ellipsoid, respectively, characterizing the wrench-generation capability of the grasp. We define the grasp-quality metric as

$$q_{\text{grasp}} = \lambda_{\min}(\mathbf{M}) \quad (1)$$

which measures the strength of wrench generation in the weakest direction. It provides a proxy for grasp quality: contact layouts with small minimum eigenvalues of $\mathbf{M}$ are more vulnerable to disturbances along certain wrench directions. We visualize this grasp quality q_{grasp} along robot hand configurations and wrench ellipsoids in 3D-moment subspace in Fig. 2.

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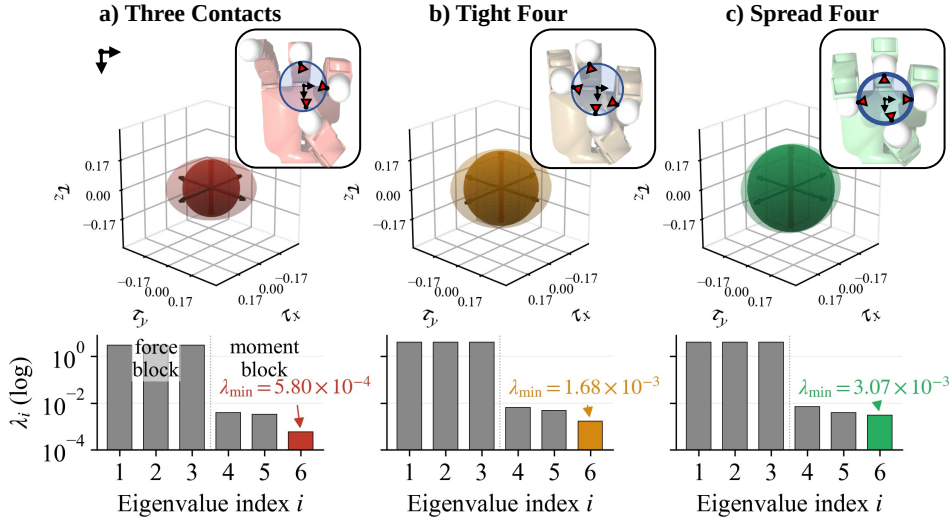


Fig. 2. Global grasp quality for three Allegro Hand configurations on a cylinder. Rows show wrench ellipsoids in the 3-D moment subspace (τ_x , τ_y , τ_z) with the contact layouts, and Gramian’s eigenvalues, with $\lambda_{\min}(\mathbf{M})$ highlighted. More evenly distributed contacts yield larger $\lambda_{\min}$, indicating better worst-case wrench resistance.

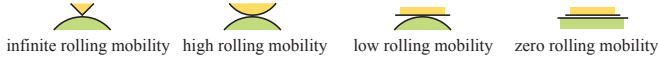


Fig. 3. Relative rolling mobility of planar bodies by rolling contact.

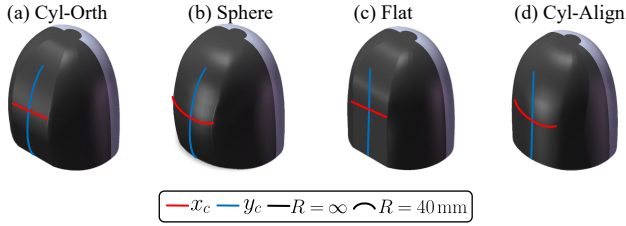


Fig. 4. Candidate fingertip morphologies.

We follow the two-stage Rapid Motor Adaptation (RMA) framework by Kumar et al. [11], adapted to dexterous manipulation by Qi et al. [6], for our training strategy. During training, we use q_{grasp} as a dense shaping term that biases the policy toward contact distributions with more balanced disturbance resistance. Because we use unit contact forces in derivation, this term can be used for policy optimization without requiring force measure at deployment.

B. Local Contact-Geometry Prior

We further analyze each finger–object contact using local contact kinematics. As illustrated in Fig. 3, at the contact interface, high relative curvature corresponds to high rotational mobility by rolling, e.g., for in-hand rolling manipulation. If rolling mobility is high, it may be difficult to precisely control rolling motion, but if rolling mobility is zero, then it is impossible to locally achieve rolling motion between the bodies. We decompose rolling mobility into task-aligned and off-axis components, promoting motion along the desired rotation direction while suppressing undesired lateral rolling.

We design and compare four fingertip morphologies: flat, spherical, task-aligned cylindrical, and task-orthogonal cylin-



Fig. 5. Experimental setup across four hand orientations and three objects.

drical, shown in Fig. 4. The flat fingertip has little rolling mobility in either directions, while the spherical fingertip yields high mobility in both directions, enabling smooth rolling but offering no directional preference. The task-aligned cylindrical fingertip preserves task-aligned mobility while suppressing off-axis rolling, whereas the task-orthogonal cylindrical fingertip exhibits the opposite behavior. Thus, the aligned cylindrical morphology induces a desired rolling direction anisotropy.

III. EXPERIMENTAL EVALUATION

We use an Allegro V4 hand mounted on a Franka Research 3 arm, as the hardware platform, as shown in Fig. 5. Before each rollout, the arm sets the palm orientation and then remains fixed; the policy does not receive this orientation information.

For each of the four fingertip designs, we train policies on cylinder objects using a teacher-student pipeline [6] in

TABLE I
TASK PERFORMANCE OF THE GRASP-QUALITY PRIOR EVALUATED ON PALM ORIENTATIONS ACROSS THREE OBJECTS.

	Tilt 0°			Tilt 45°			Tilt 90°			Tilt 180°			Avg		
	Rot	TTT	SR	Rot	TTT	SR	Rot	TTT	SR	Rot	TTT	SR	Rot	TTT	SR
Default + Vanilla	0.97	17.4	30%	0.70	19.2	33%	0.43	16.7	15%	1.02	13.0	19%	0.78	16.6	24%
Default + GQ	2.27	27.6	78%	1.04	27.0	67%	0.57	19.6	37%	1.13	16.5	44%	1.25	22.7	56%
Aligned + Vanilla	1.59	33.4	81%	1.09	29.5	78%	0.59	26.5	48%	0.99	15.1	48%	1.06	26.1	64%
Aligned + GQ	1.83	33.6	96%	1.27	32.9	81%	0.67	32.5	81%	1.32	25.4	74%	1.27	31.1	83%

TABLE II
TASK PERFORMANCE OF FINGERTIP DESIGNS EVALUATED ON OBJECTS ACROSS FOUR PALM ORIENTATIONS.

Fingertip	Cylinder			Cuboid			Ball			Avg		
	Rot	TTT	SR	Rot	TTT	SR	Rot	TTT	SR	Rot	TTT	SR
Aligned	2.29	31.9	86%	0.69	29.9	83%	0.83	31.6	81%	1.27	31.1	83%
Orth	0.78	21.6	67%	0.20	17.6	25%	1.08	14.3	36%	0.68	17.8	43%
Spherical	1.39	25.1	86%	0.33	20.3	39%	1.01	16.3	34%	0.91	20.6	53%
Flat	1.46	23.3	69%	0.42	26.9	50%	0.67	12.6	28%	0.85	20.9	49%

Isaac Gym [12] with and without the grasp-quality shaping, and evaluate them on hardware. We report mean cumulative rotation (**Rot**), mean episode duration capped at 35 s (**TTT**), and gated success rate (**SR**). Experiments span three objects and four palm orientations, as shown in Table I and II.

A. Evaluation on the Global Grasp-Quality Prior

We use the default Allegro Hand’s fingertip and the aligned cylindrical fingertip as the basis to evaluate the effect of the global grasp-quality (GQ) prior. Across three objects and four palm tilts, the GQ term improves both rotation efficiency and stability (Table I): overall SR increases from 24% to 56% with the default fingertip, and from 64% to 83% with the aligned fingertip. TTT also increases from 16.6 to 22.7s and from 26.1 to 31.1s, while Rot improves from 0.78 to 1.25 and from 1.06 to 1.27, respectively.

B. Evaluation on the Local Contact-Geometry Prior

We evaluate the effect of fingertip design using the policy trained with the grasp-quality term (Table II). On the cylinder object used in training, the aligned cylindrical fingertip is the most productive and stable design, achieving the largest rotation (2.29 turns) and longest holding time (31.9s). The spherical fingertip holds nearly as long but rotates much less (1.39 turns), suggesting that it permits rolling without sufficiently constraining motion to the task direction. The orthogonal fingertip performs worst across all metrics in average, with the flat fingertip showing comparable results. The aligned design achieves the highest TTT and SR on all three objects. Its Rot is slightly lower than that of the orthogonal and spherical fingertips on the ball, whose geometry is less matched with the trained cylinder objects.

IV. LIMITATIONS AND CONCLUSIONS

Our framework does not explicitly parameterize coordinated fingertip motion in a geometric task space; instead, coordination emerges from the learned policy, while the proposed priors improve grasp robustness and local rolling

behavior. A complementary direction is to integrate cooperative geometric primitives [13] as a higher-level coordination representation. Their analytical Jacobians could guide coordinated fingertip motion, with our grasp-quality metric serving as a secondary objective for maintaining well-conditioned contacts and the anisotropic fingertip morphology passively regulating local rolling. Such an integration could combine explicit task-space coordination with physics-informed robustness, although its effectiveness under contact transitions and uncertain contact states remains to be investigated.

This work uses in-hand rotation to study how task-relevant physics priors can improve robotic skill learning. We integrate complementary global grasp-quality and local contact-geometry priors into reinforcement learning and fingertip design, demonstrating that jointly embedding contact information in learning and morphology provides effective inductive biases for dexterous manipulation.

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